

SAT Math Mastery · Week TP1L**SAT Trick Pack 1 — Part L****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** shifted-exponent rewriting · similar solids · sectors · vertex-form symmetry · radicals as powers · growth multipliers and logs · break-even inequalities · altitude in a circle · shifted quadratics · inscribed polygons · feasible regions · data transformations · decay to a floor**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Rewrite mixed bases over one base and read the coefficient as the value at the shifted input.
- S2.** Turn an area (or volume) ratio into the linear factor k , then apply k^2 or k^3 to the other quantity.
- S3.** Obtain a sector's fraction from an area ratio and apply it to the circumference.
- S4.** Use vertex form: two inputs with the same output sit symmetrically about the axis.
- S5.** Write related radicals as powers of one base, find the base from the given product, then evaluate the target.
- S6.** Convert a percent increase over a period into a full multiplier and solve for time with a logarithm.
- S7.** Compare two linear plans, find the break-even, and step to the correct whole number.
- S8.** Use the right angle in a semicircle and the altitude to the hypotenuse to get ratios.
- S9.** Undo an input shift to move a vertex, then use the vertex as the midpoint of the two zeros.
- S10.** Express inscribed polygons through the circumradius and subtract areas.
- S11.** Maximise a linear objective over a polygonal region at its vertices; treat strict inequalities with care.
- S12.** Apply $y = ax + b$ to centre and spread separately.
- S13.** Read a model with a vertical shift: the base multiplies the gap to the asymptote.

2 · REQUIRED FACTS

- $a^m a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $\sqrt[n]{a} = a^{1/n}$, $(ab)^x = a^x b^x$; at the input that makes the exponent 0, a power equals 1.
- Similar solids with linear factor k : lengths $\times k$, areas $\times k^2$, volumes $\times k^3$.
- A sector of angle θ is the fraction $\frac{\theta}{2\pi}$ of the circle; its area ratio to the circle equals its arc ratio to the circumference.
- $a(x-p)^2 + q$ has axis $x = p$; equal outputs at x_1, x_2 mean $p = \frac{x_1 + x_2}{2}$ and $x_{1,2} = p \pm d$.
- An angle inscribed in a semicircle is a right angle; an equilateral triangle inscribed in a circle of radius r has side $r\sqrt{3}$; an inscribed square has diagonal $2r$.
- A $p\%$ increase per period multiplies by $1 + \frac{p}{100}$; $A b^t = c \iff t = \log_b \frac{c}{A}$.
- For data: $y = ax + b$ gives $\text{mean}' = a \cdot \text{mean} + b$ and $\text{SD}' = |a| \cdot \text{SD}$ (same for IQR and range).
- $T = L + A b^t$ approaches the level L ; $T - L$ is multiplied by b each step.

3 · REQUIRED FORMULAS

$$CD^2 = AD \cdot DB, \quad \frac{V_2}{V_1} = k^3, \quad \frac{S_2}{S_1} = k^2, \quad \ell = \frac{\theta}{2\pi} \cdot 2\pi r = r\theta, \quad A_{\text{sector}} = \frac{1}{2}r^2\theta$$

$$f(x) = a(x-p)^2 + q, \quad x_{1,2} = p \pm \sqrt{\frac{q-y}{-a}} \text{ (height } y), \quad A_{\Delta} = \frac{3\sqrt{3}}{4}r^2, \quad A_{\square} = 2r^2$$

4 · MINI THEORY SHEET

Shifted exponents. Break every base into primes, collect each prime's exponent, and compare with the target form $k b^{x-s}$. Evaluate at $x = s$: the power is 1, so k is the value of the function there, and b is the ratio of consecutive values.

Scaling. Never apply an area ratio to a volume. First recover k (square root of the area ratio, cube root of the volume ratio), then use k , k^2 or k^3 for the quantity asked.

Symmetry. For a parabola $a(x-p)^2 + q$, the set where the output is at least y is an interval centred at p ; its length is $2\sqrt{(q-y)/|a|}$ and does not depend on p .

Break-even. Solve the equality, decide which plan has the smaller rate, and move to the next whole number on the side where the stated plan is strictly cheaper.

Feasible regions. Linear objectives are extreme at vertices. List the vertices, evaluate, compare. A strict inequality " $> k$ " has no solution when $k \geq \max$.

Vertical shift. For $L + Ab^t$, subtract L first; only then does b act as a multiplier.

Speed shortcuts.

Situation	Shortcut
mixed bases, answer is a coefficient	evaluate at the input that makes the exponent 0
areas given, volume wanted	$k = \sqrt{\text{area ratio}}$, then k^3 (and reverse with a cube root)
sector area and circle area	fraction = area ratio; multiply by the circumference
height reached at two times	$p \pm d$ with $16d^2 = q - y$; the gap is $2d$
radicals of one variable	$\sqrt[n]{m} = m^{1/n}$; find m from the product, then evaluate
$p\%$ per period	multiplier $1 + \frac{p}{100}$ per period; divide by the start value to remove the constant
two plans, whole numbers	solve the equality, then jump to the next integer on the cheaper side
altitude to the hypotenuse in a circle	$CD^2 = AD \cdot DB$; leg ratio = $\sqrt{\text{segment ratio}}$
$f(x+s)$ has vertex at v	f has vertex at $v+s$; other zero = $2(v+s)$ - known zero
inscribed square and triangle	square diagonal = $2r$; triangle side = $r\sqrt{3}$
objective $> k$ over a polygon	$k \geq \max$ at the vertices
$y = ax + b$ on data	centre: $a \cdot c + b$; spread: $ a \cdot s$
$L + Ab^t$	subtract L ; b is the factor on the gap

5 · GUIDED EXAMPLES

Example 1 — Coefficient at the shifted input

$h(x) = \frac{2^{x+1} \cdot 9^x}{3^{x-1}}$ can be written as $k \cdot 6^{x-2}$. Find k .

Evaluate where the exponent of 6 is 0, i.e. $x = 2$: $k = h(2) = \frac{2^3 \cdot 81}{3^1} = 216$.

(Check: $h(x) = 2 \cdot 3 \cdot \frac{2^x 3^{2x}}{3^x} = 6 \cdot 6^x = 216 \cdot 6^{x-2}$.)

Example 2 — Two times at one height

A jet of water has height $h(t) = -16t^2 + bt + c$ with maximum 52 ft. For how long is it at least 36 ft high?

$h = -16(t-p)^2 + 52 \geq 36 \Rightarrow (t-p)^2 \leq \frac{16}{16} \Rightarrow |t-p| \leq 1$. Duration = $2 \cdot 1 = 2$ seconds.

Example 3 — Objective over a region

Region: $x \geq 0$, $y \geq 0$, $x + y \leq 10$, $3x + y \leq 18$. Greatest value of $2x + y$?

Vertices: $(0,0)$, $(6,0)$, $(0,10)$, $(4,6)$. Values: 0, 12, 10, 14. The greatest is 14.

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = \frac{6^{x+1} \cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x) = a \cdot b^{x-1}$, where a and b are positive constants. What is the value of $a + b$?

(A) 62

(B) 440 ✓

(C) 456

(D) 3456

1. Evaluate at $x = 1$, where $b^{x-1} = 1$: $a = f(1) = \frac{6^2 \cdot 4}{3^{-1}} = 36 \cdot 4 \cdot 3 = 432$.

2. Find b from the per-unit multiplier: $\frac{f(x+1)}{f(x)} = \frac{6 \cdot 4}{3} = 8$, so $b = 8$.

3. Check: $f(x) = 6 \cdot 9 \cdot \frac{6^x 4^x}{3^x} = 54 \cdot 8^x = 432 \cdot 8^{x-1}$.

4. $a + b = 432 + 8 = 440$.

Answer: (B)

2.

SAT CHALLENGE · CALCULATOR

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 7.5

(B) 22.5

(C) 67.5 ✓

(D) 202.5

1. Sealant is proportional to surface area, so the area ratio is $\frac{14.4}{1.6} = 9 = k^2$.

2. The linear scale factor is $k = \sqrt{9} = 3$.

3. Volumes scale by $k^3 = 27$: $2.5 \cdot 27 = 67.5$.

4. (Using the area ratio 9 directly would give 22.5, the wrong quantity.)

Answer: (C)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

(A) $30 + 6\pi$ ✓(B) $15 + 6\pi$ (C) $30 + 3\pi$ (D) $30 + 45\pi$

1. Area of the whole face: $\pi(15)^2 = 225\pi$. The sector is $\frac{45\pi}{225\pi} = \frac{1}{5}$ of the circle.
2. Circumference: $2\pi(15) = 30\pi$, so the arc length is $\frac{1}{5} \cdot 30\pi = 6\pi$.
3. Perimeter of the sector = arc + two radii = $6\pi + 2(15) = 30 + 6\pi$.
4. (Check with $A = \frac{1}{2}r^2\theta$: $45\pi = \frac{1}{2}(225)\theta \Rightarrow \theta = \frac{2\pi}{5}$, and arc = $r\theta = 6\pi$.)

Answer: (A)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

A rocket's height, in feet, t seconds after launch is modeled by $h(t) = -16t^2 + bt + c$, where b and c are constants. The rocket is launched from a height of at most 94 feet, and the maximum height it reaches is 130 feet. For how many seconds is the rocket at a height of at least 94 feet?

1. Write the function in vertex form: $h(t) = -16(t - p)^2 + 130$, where p is the time of the maximum.
2. Height at least 94: $-16(t - p)^2 + 130 \geq 94 \Rightarrow (t - p)^2 \leq \frac{36}{16} = \frac{9}{4}$.
3. So $|t - p| \leq \frac{3}{2}$, i.e. $p - \frac{3}{2} \leq t \leq p + \frac{3}{2}$. The value of p does not matter (launch height at most 94 keeps the whole interval after $t = 0$).
4. Duration = $2 \cdot \frac{3}{2} = 3$ seconds. (Check with $p = \frac{5}{2}$: $h(1) = h(4) = 94$.)

Answer: 3

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive real number m , let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2 ?

1. Write both with the same base: $a = m^{1/3}$, $b = m^{1/4}$.
2. $ab = m^{1/3+1/4} = m^{7/12} = 128 = 2^7$, so $m = 2^{12} = 4096$ (the function $m^{7/12}$ is increasing, so this is the only solution).
3. $ab^2 = m^{1/3+1/2} = m^{5/6} = 2^{10} = 1024$.
4. (Check: $a = 16$, $b = 8$, $ab = 128$, $ab^2 = 16 \cdot 64 = 1024$.)

Answer: 1024

6.

SAT CHALLENGE · CALCULATOR

An investment grows by 6% every half-year. If V_0 is its initial value, the value after t years is $V(t)$. Which of the following gives the number of years t until the value is three times the initial value?

(A) $t = 2 \log_{1.06} 3$

(B) $t = \log_{1.12} 3$

(C) $t = \log_{1.06} \frac{3}{2}$

(D) $t = \frac{1}{2} \log_{1.06} 3$ ✓

1. A 6% increase multiplies the value by 1.06 every half-year, so in t years there are $2t$ periods: $V(t) = V_0(1.06)^{2t}$.
2. Set $V(t) = 3V_0$ and divide by V_0 to cancel the constant: $(1.06)^{2t} = 3$.
3. Take $\log_{1.06}$ of both sides: $2t = \log_{1.06} 3$, so $t = \frac{1}{2} \log_{1.06} 3$.
4. (Numerically $t \approx 9.43$ years; $(1.06)^{18.86} \approx 3$.)

Answer: (D)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

1. Volumes scale by k^3 : $k^3 = \frac{432}{250} = \frac{216}{125}$, so $k = \sqrt[3]{\frac{216}{125}} = \frac{6}{5}$.

2. Glass area scales by $k^2 = \frac{36}{25}$.

3. $400 \cdot \frac{36}{25} = 576$ square decimeters.

Answer: 576

8.

SAT CHALLENGE · CALCULATOR

Plan A charges a one-time enrollment fee of \$85 plus \$18.50 per month. Plan B has no enrollment fee and charges \$28 per month. For a whole number of months m , which of the following describes all values of m for which Plan A costs strictly less than Plan B in total?

(A) $m \geq 10$

(B) $m \geq 8$

(C) $m \geq 9$ ✓

(D) $m \leq 8$

1. Plan A is cheaper when $85 + 18.5m < 28m$, i.e. $85 < 9.5m$.

2. $m > \frac{85}{9.5} \approx 8.95$.

3. Plan A has the smaller monthly rate, so it becomes cheaper only after the break-even. Since m must be a whole number, the smallest valid value is 9: $m \geq 9$.

4. (Check: $m = 8$: A = 233, B = 224, so B is cheaper. $m = 9$: A = 251.5, B = 252, so A is cheaper.)

Answer: (C)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the value of $\frac{AC}{CB}$?

(A) $\frac{2}{5}$

(B) $\frac{1}{2}$ ✓

(C) $\frac{4}{5}$

(D) $\frac{1}{4}$

1. C lies on a circle with diameter AB , so $\angle ACB = 90^\circ$ and $\overline{CD}$ is the altitude to the hypotenuse.
2. $CD^2 = AD \cdot DB$ and $AD + DB = 20$: $AD \cdot DB = 64$. Solving $x(20 - x) = 64$ gives $x = 4$ or 16 ; with $AD < DB$, $AD = 4$ and $DB = 16$.
3. $AC^2 = AD \cdot AB = 4 \cdot 20 = 80$ and $CB^2 = DB \cdot AB = 16 \cdot 20 = 320$.
4. $\frac{AC}{CB} = \sqrt{\frac{80}{320}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$. (Note $\frac{AD}{DB} = \frac{1}{4}$ is the ratio of the segments, not of the legs.)

Answer: (B)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $f(x) = 2x^2 + bx + c$ has $(x - 9)$ as a factor. The function g is defined by $g(x) = f(x + 3)$, and the graph of g in the xy -plane has its vertex at $x = 1$. What is the other zero of f ?

1. The graph of $g(x) = f(x + 3)$ is the graph of f shifted 3 units left. So the vertex of f is at $x = 1 + 3 = 4$.
2. The vertex lies midway between the zeros of f . One zero is 9, so the other is $2(4) - 9 = -1$.
3. (Check: $-\frac{b}{4} = 4 \Rightarrow b = -16$; $f(9) = 0 \Rightarrow c = -162 + 144 = -18$; $f(x) = 2(x - 9)(x + 1)$ and g has vertex at $x = 4 - 3 = 1$.)

Answer: -1

11.

SAT CHALLENGE · CALCULATOR

A circle has area 48π . A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

(A) $96 - 12\sqrt{3}$

(B) $192 - 36\sqrt{3}$

(C) $96 - 27\sqrt{3}$

(D) $96 - 36\sqrt{3}$ ✓

1. $\pi r^2 = 48\pi$, so $r^2 = 48$ and $r = 4\sqrt{3}$.

2. Square: its diagonal is the diameter, $2r = 8\sqrt{3}$. Area = $\frac{d^2}{2} = \frac{192}{2} = 96$.

3. Triangle: the side is $r\sqrt{3} = 12$, so the area is $\frac{\sqrt{3}}{4} \cdot 12^2 = 36\sqrt{3}$.

4. Difference: $96 - 36\sqrt{3}$.

Answer: (D)

12.

SAT CHALLENGE · DESMOS

$$x \geq 0, \quad y \geq 0, \quad 2x + y \leq 10, \quad x + 3y \leq 15, \quad 3x + 2y > k$$

In the system of inequalities above, k is a constant. What is the least value of k for which the system has no solution (x, y) ?

1. The first four inequalities describe a polygon. Its vertices are $(0, 0)$, $(5, 0)$, $(0, 5)$ and the intersection of $2x + y = 10$ and $x + 3y = 15$, which is $(3, 4)$.

2. Evaluate $3x + 2y$ at the vertices: 0, 15, 10 and $9 + 8 = 17$. The greatest value over the region is 17.

3. $3x + 2y > k$ has no solution in the region exactly when $k \geq 17$ (for $k = 17$ the strict inequality fails everywhere, since $3x + 2y \leq 17$).

4. The least such k is 17.

Answer: 17

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

(A) -4 ✓

(B) 3

(C) 11

(D) -18

1. Median transforms like every value: $15 - \frac{52}{2} = -11$.

2. Spread (IQR) is multiplied by the absolute value of the multiplier and is not changed by the added constant: $\left| -\frac{1}{2} \right| \cdot 14 = 7$.

3. Sum: $-11 + 7 = -4$.

4. (Check with the data 30, 44, 46, 50, 54, 58, 60, 70: median 52, IQR $59 - 45 = 14$; new values give median -11 , quartiles -14.5 and -7.5 , IQR 7.)

Answer: (A)

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

(A) T falls 15% per minute(B) the gap $T - 24$ falls 15% per minute ✓(C) $T(1) = 0.85T(0)$ (D) T levels off at 85% of $T(0)$

1. The constant 24 is the room temperature: the tea cools toward 24°C , not toward 0.

2. Subtract it: $T(t) - 24 = 66(0.85)^t$. Each minute this difference is multiplied by 0.85, so it decreases by 15%.

3. The tea's own temperature does not decrease by 15% per minute: $T(0) = 90$ and $T(1) = 80.1$, a drop of about 11%.

4. So 0.85 is the factor by which the difference between the tea and the room changes each minute.

Answer: (B)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all real numbers x , the function g is defined by $g(x) = \frac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k ?

1. Because $15^{x+1} = 1$ when $x = -1$, the constant k equals $g(-1)$.

2. $g(-1) = \frac{9^1 \cdot 10^0}{6^{-2}} = 9 \cdot 36 = 324$.

3. (Check by primes: $g(x) = \frac{3^{2x+4} 2^{x+1} 5^{x+1}}{2^{x-1} 3^{x-1}} = 3^{x+5} \cdot 2^2 \cdot 5^{x+1} = 4 \cdot 3^5 \cdot 5 \cdot 15^x = 4860 \cdot 15^x = 324 \cdot 15^{x+1}$.)

Answer: 324